rise up through water or any fluid. This is also the reason for the clouds in the sky to move in the upward direction.

Point to ponder

i) The terminal speed of a sphere is directly proportional to the square of the radius of the sphere. Hence, larger raindrops fall with greater speed as compared to the smaller raindrops.
ii) If the density of the material of the sphere is less than the density of the medium, then the sphere shall attain terminal velocity in the upward direction. That is why gas bubbles rise up in soda water.

7.4.6 Stoke's law and its applications

When a body falls through a viscous medium, it drags the layer of the fluid immediately in contact with it. This produces a relative motion between the different layers of the liquid. Stoke performed many experiments on the motion of small spherical bodies in different fluids and concluded that the viscous force F acting on a spherical body of radius r depends directly on

i) radius (r) of the sphere
ii) velocity (υ) of the sphere and
iii) coefficient of viscosity η of the liquid

Therefore $F \propto \eta^x r^y \upsilon^z \Rightarrow F = k\eta^x r^y \upsilon^z$, where k is a dimensionless constant.

Using dimensions, the above equation can be written as

$$\left[MLT^{-2}\right] = k\left[ML^{-1}T^{-1}\right]^x \times \left[L\right]^y \times \left[LT^{-1}\right]^z$$

On solving, we get x=1, y=1, and z=1

Therefore, $F=k\eta\, r\upsilon$

Experimentally, Stoke found that the value of $k = 6\pi$

$$F = 6\pi\eta\, r\upsilon \qquad (7.23)$$

This relation is known as Stoke's law

Practical applications of Stoke's law

Since the raindrops are smaller in size and their terminal velocities are small, remain suspended in air in the form of clouds. As they grow up in size, their terminal velocities increase and they start falling in the form of rain.

This law explains the following:

a) Floatation of clouds
b) Larger raindrops hurt us more than the smaller ones
c) A man coming down with the help of a parachute acquires constant terminal velocity.

7.4.7 Poiseuille's equation

Poiseuille analyzed the steady flow of liquid through a capillary tube. He derived an expression for the volume of the liquid flowing per second through the capillary tube.

As per the theory, the following conditions must be retained while deriving the equation.

- The flow of liquid through the tube is streamlined.
- The tube is horizontal so that gravity does not influence the flow
- The layer in contact with the wall of the tube is at rest
- The pressure is uniform over any cross section of the tube

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